Graphics

National 5 · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain how graphics are used to generate, explore, refine and communicate ideas
- ☐ choose a suitable graphic for a purpose: orthographic, sectional, exploded, pictorial, working drawing, render
- ☐ describe the advantages of manual and computer-generated graphics
- ☐ describe what a working drawing shows

I DO — Worked example

Read how a full-mark answer is built

1 Outline two pieces of information shown on a working drawing. (2 marks)

Full-mark answer each ✓ is one mark

All the dimensions needed to make each part ✓. Orthographic views of every component with the material each is made from ✓.

Why marks are not awarded

"It shows what the product looks like and how it works."

0 marks: both responses were credited 0 in SQA marking.

WE DO — Guided practice

Use the starters and word bank to help you

A designer presents a speaker as a computer-generated render.

2 Describe two benefits of using computer-generated graphics. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

proportion · realistically · aesthetics · CAM · colour · client · texture · file

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which graphic communicates dimensions most clearly?
☐ Rendered illustration ☐ Orthographic drawing ☐ Mood board ☐ Freehand thumbnail

b) Which is an advantage of CAD over manual drawing?
☐ No training needed ☐ Changes can be made without redrawing ☐ Cheaper equipment ☐ Better at very rough early ideas

c) Which graphic best shows the inside of a product?
☐ Exploded view ☐ Sectional view ☐ One-point perspective ☐ Mood board

d) Why would a designer show a client a rendered 3D illustration rather than an orthographic drawing?
☐ It includes all dimensions ☐ Non-experts find it easier to understand ☐ It is quicker than sketching ☐ It is needed for CNC

4 Key terms: write the letter of the matching description next to each term.

1. **Sectional view**	____	**A** Shows how components fit together.
2. **Orthographic drawing**	____	**B** Shows the interior detail of a product.
3. **Exploded view**	____	**C** Shows dimensions and proportions in 2D views (elevation, end elevation, plan).
4. **Working drawing**	____	**D** Gives the manufacturer all views, dimensions and technical details needed.

5 Describe the advantage of manual sketching. (2 marks)

6 Describe the advantage of CAD graphics. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?